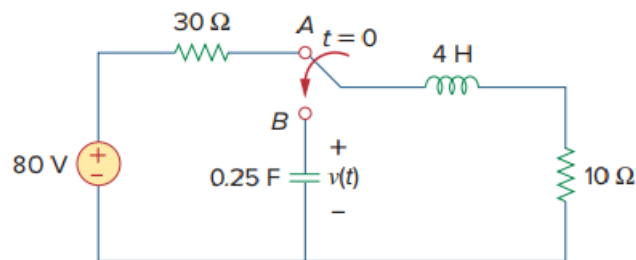


Problem

The switch in Fig. 8.69 moves from position A to position B at $t = 0$ (please note that the switch must connect to point B before it breaks the connection at A, a make-before-break switch). Let $v(0) = 0$, find $v(t)$ for $t > 0$.

Figure 8.69:



Step-by-step solution

Step 1 of 9

Refer to Figure 8.69 in the textbook.

Re-draw the circuit diagram.

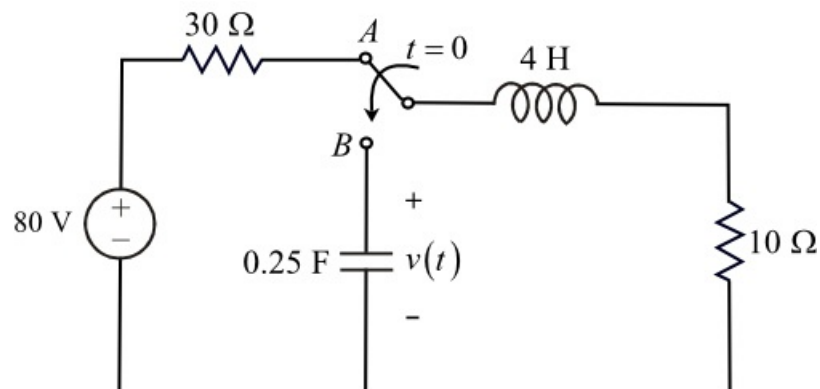


Figure 1

Step 2 of 9

At $t = 0^-$ the switch is connected to position 'A'. The inductor acts like a short circuit, while the capacitor acts like an open circuit. The equivalent circuit is shown in Figure 2.

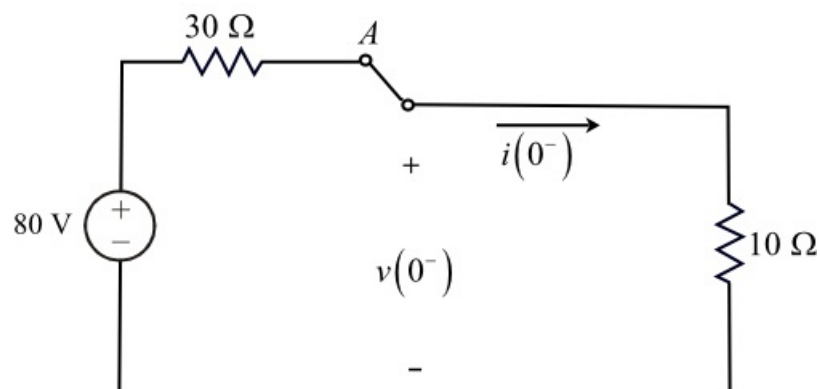


Figure 2

Step 3 of 9

Calculate the initial current using Ohm's law.

$$\begin{aligned} i(0^-) &= \frac{80}{30+10} \\ &= 2 \text{ A} \end{aligned}$$

Since current through inductor cannot change instantaneously.

$$\begin{aligned} i(0) &= i(0^-) \\ &= 2 \text{ A} \end{aligned}$$

Step 4 of 9

At $t = 0^+$ the switch is connected to position 'B'. Draw the modified circuit diagram.

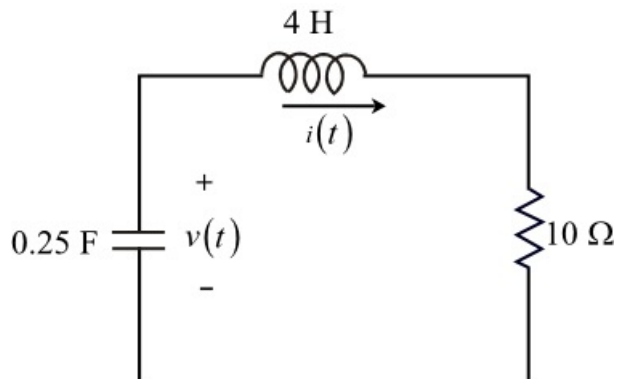


Figure 3

Step 5 of 9

Calculate the neper frequency.

$$\begin{aligned} \alpha &= \frac{R}{2L} \\ &= \frac{10}{(2)(4)} \\ &= 1.25 \end{aligned}$$

Calculate the resonant frequency.

$$\begin{aligned} \omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{0.25 \times 4}} \\ &= 1 \end{aligned}$$

Calculate the roots of characteristic equation.

$$\begin{aligned} s_{1,2} &= -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \\ &= -1.25 \pm \sqrt{1.25^2 - 1} \\ &= -1.25 \pm 0.75 \\ &= -2, -0.5 \end{aligned}$$

Step 6 of 9

Since $\alpha > \omega_0$ the response is over damped.

Therefore, the response is,

$$v(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \text{ V}$$

$$v(t) = A_1 e^{-0.5t} + A_2 e^{-2t} \text{ V} \dots\dots (1)$$

Consider the initial voltage across the capacitor is, $v(0) = 0 \text{ V}$.

Substitute 0 for t in equation (1).

$$v(0) = A_1 e^{-0.5(0)} + A_2 e^{-2(0)} \text{ V}$$

$$v(0) = A_1 + A_2$$

$$0 = A_1 + A_2$$

$$A_1 = -A_2 \dots\dots (2)$$

Step 7 of 9

Write the expression for voltage across the capacitor.

$$v(t) = \frac{1}{C} \int i(t) dt \dots\dots (3)$$

Differentiate equation (3) with respect to ' t '.

$$\frac{dv(t)}{dt} = \frac{1}{C} i(t) \dots\dots (4)$$

Substitute 0 for t in equation (4).

$$\begin{aligned} \frac{dv(0)}{dt} &= \frac{1}{C} i(0) \\ &= \frac{1}{(0.25)} (2) \\ &= 8 \text{ V/s} \end{aligned}$$

Step 8 of 9

Differentiate equation (1) with respect to ' t '.

$$\frac{dv(t)}{dt} = -0.5 A_1 e^{-0.5t} - 2 A_2 e^{-2t}$$

$$\frac{dv(0)}{dt} = -0.5 A_1 - 2 A_2$$

$$-0.5 A_1 - 2 A_2 = 8 \dots\dots (5)$$

Substitute $-A_2$ for A_1 in the equation (5).

$$-0.5(-A_2) - 2 A_2 = 8$$

$$0.5 A_2 - 2 A_2 = 8$$

$$-1.5 A_2 = 8$$

$$A_2 = -5.33$$

Substitute -5.33 for A_2 in equation (2).

$$\begin{aligned} A_1 &= -(-5.33) \\ &= 5.33 \end{aligned}$$

Step 9 of 9

Substitute the values of A_1 and A_2 in equation $v(t) = A_1 e^{-0.5t} + A_2 e^{-2t}$ V.

$$v(t) = 5.33e^{-0.5t} - 5.33e^{-2t} \text{ V}$$

Hence, the voltage $v(t)$ is $\boxed{5.33e^{-0.5t} - 5.33e^{-2t} \text{ V}}$.